

Hermite porous-fin solver

Comparison with the governing equations in the supplied paper

Technical discussion note | 8 October 2026

Scope. The current implementation solves the printed dimensionless equation, Eq. (13), with an explicit interpretation of Eq. (14). It uses a different numerical method and illustrative dimensionless coefficients. It does not yet reproduce the paper’s calibrated SiC or Si₃N₄ cases.

1. Equation preserved by the implementation

$$(1 - X^2)\theta_\tau = \frac{\partial}{\partial X}[(1 - X^2)\theta_X] - \sqrt{1 - X^2} F(\theta)$$
$$F(\theta) = N_c\theta^2 + N_r\theta + n_1^2\theta^{p+1} + n_0^2n_3\theta^p$$

This is algebraically equivalent to Eq. (13): differentiating the flux gives $(1 - X^2)\theta_{XX} - 2X\theta_X$. All four printed nonlinear/loss terms and their signs are retained. The signed moisture coefficient $n_0^2n_3$ is not replaced by its absolute value.

Data: $0 \leq X \leq 1$; $\theta(X,0)=0$ in the interior; $\theta(0,\tau)=1$ after the base step; $\theta_X(1,\tau)=0$. Fin efficiency is evaluated using dimensionless Eq. (16).

2. What differs from the paper

Aspect	Paper	Current implementation
Space approximation	Finite differences	Piecewise shifted Hermite polynomials; weighted Galerkin formulation
Time integration	Described as an implicit finite-difference scheme	Implicit BDF integration with analytic Jacobian
Coefficients	Material and environmental quantities mapped to dimensionless groups	Explicit illustrative dimensionless presets
Validation	Comparison with published HAM results	Manufactured solution, refinement and independent finite-volume calculation

Changing the numerical method does not change Eq. (13). The coefficient presets do change the particular problem being solved: their values have not been calculated from the paper’s ceramic-material baseline.

Source and review basis

P. L. Pavan Kumar, B. J. Gireesha and P. Venkatesh, “Ceramic-based semi-spherical porous fin under condensation condition: a transient study on thermal performance and efficiency”, International Journal of Mechanics and Materials in Design 22:72 (2026). DOI: [10.1007/s10999-026-09893-6](https://doi.org/10.1007/s10999-026-09893-6). Equation numbering and printed PDF pages 7–10 are used below.

This note reports a reading of the printed equations and the local implementation. Apparent printing or derivation errors require confirmation of the authors’ intended formulation.

Source inconsistencies and choices

The following observations distinguish direct contradictions from interpretation and algebraic inference.

3. Dimensional initial condition versus dimensionless data

Eq. (11) prints $T(x,0)=0$. With $\theta=(T_a-T)/(T_a-T_b)$, this gives $\theta(X,0)=T_a/(T_a-T_b)$, rather than the zero value printed in Eq. (14). Zero θ would require an initial dimensional temperature $T=T_a$.

Choice: follow dimensionless Eq. (14); the intended dimensional initial temperature remains unresolved.

4. Starting-corner incompatibility

Eq. (14) prescribes $\theta(X,0)=0$ and $\theta(0,\tau)=1$, assigning different values at $X=0$, $\tau=0$. This is an incompatibility for smooth data, but a common interpretation is an abruptly imposed boundary temperature. **Treatment:** use a zero initial interior state and a base step to 1 at time zero. The numerical startup approximation and its limitations are explicit on the next page.

5. Radiation factor in Eq. (10)

Eq. (7) linearizes the radiation difference as $T_a^4 - T^4 \approx 4T_a^3(T_a - T)$. Accordingly, the radiation term derived from Eq. (1) contains T_a^3 . The printed Eq. (10) omits this factor, while N_r in Eq. (12) includes it. **Choice:** follow the $N_r \theta$ term in Eq. (13), with N_r supplied directly. The implementation does not resolve the dimensional derivation discrepancy.

6. Apparent factor-of-two mismatch in N_c

Eq. (10) contains a leading factor 2 in its Darcy/convection term. Under $X=x/R$ and the printed definition of θ , its dimensionless coefficient becomes:

$$\frac{2\rho g K \beta C_p R (T_a - T_b)}{k_{\text{eff}} \nu_f} = 2N_c$$

Here N_c is the definition in Eq. (12). Direct substitution therefore appears to give $-2N_c \sqrt{(1-X^2)\theta^2}$, whereas Eq. (13) prints $-N_c \sqrt{(1-X^2)\theta^2}$. **Choice:** retain Eq. (13) as printed. Confirmation is needed as to whether Eq. (12)'s definition or Eq. (13)'s coefficient should change.

7. Table 3 versus the base boundary condition

Table 3 labels its temperature samples as $X=0$ and $\tau=0.5$, but lists values around 0.8455. Eq. (14) requires $\theta(0,\tau)=1$. **Treatment:** enforce the boundary condition and do not use that table entry as a validation target. A mistaken coordinate label or another unreported convention could explain it; the intended correction is unknown.

Interpretation: these choices make one stated dimensionless problem computable. They do not establish that every dimensional equation, table and physical interpretation in the paper is mutually consistent.

Numerical handling and evidence

8. Degenerate tip: a numerical issue, not a contradiction

At $X=1$, $A(X)=1-X^2$ vanishes. Dividing the PDE by A introduces singular coefficients. The solver instead keeps the conservative equation and integrates it against Hermite test functions. It imposes the base value and zero tip derivative through linear constraints on the basis, with continuous values and slopes between cells.

$$M\dot{c} + Kc + \int_0^1 \sqrt{1-X^2} v F(1+Vc) dX = 0$$

The reconstructed temperature is $\theta=1+Vc$. The mass matrix uses weight A , and the stiffness matrix represents the conductive flux. Quadrature nodes stay inside the cells, avoiding direct evaluation at the degenerate endpoint. This is an alternative numerical formulation of the printed dimensionless PDE.

9. What the startup treatment actually does

The integrator starts from a weighted L^2 projection of the zero interior state into the constrained Hermite space. A finite polynomial representation cannot reproduce the discontinuous boundary step exactly. The default weighted initial projection error is approximately **0.0227**; this is a projection diagnostic, not a later-time solution error.

The saved time-zero display frame is explicitly set to the prescribed data (zero interior, base one). It is not the polynomial reconstruction of the initial coefficients. All later frames come from the numerical solution. Residual and comparison acceptance checks begin at $\tau=0.05$; earlier startup accuracy has not been certified.

10. Current validation results

Default case / diagnostic	Measured value
$N_c=0.5$; $N_r=0.1$; $n_1^2=0.5$; $n_0^2 n_3=0.1$; $p=1$	12 cells; 6 Hermite functions/cell; final $\tau=0.8$
Maximum difference from independent 400-cell finite-volume solution, after startup	3.58e-06
Independent weak residual	1.09e-06
Sampled weighted strong PDE residual	4.56e-03
Base value / tip derivative errors	2.22e-16 / 1.04e-12

Refinement reduces the sampled strong residual. Tests also compare against an 800-cell reference and a manufactured solution with analytically derived forcing. The finite-volume comparison is numerical, not exact. Agreement supports the implemented problem; it does not prove the authors' intended physical model or provide a rigorous error bound.

Questions for academic review

1. Should the initial dimensional temperature be T_a , or should Eq. (14) change?
2. Which N_c convention resolves the factor-of-two mismatch?
3. Which coordinate was intended in Table 3?
4. Which consistent physical inputs and property conventions should be used to calibrate the SiC and Si₃N₄ cases?